BIBI ZAINAB

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ECE GRADUATE | PYTHON | MACHINE LEARNING | AI AGENTS | CYBERSECURITY

PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with hands-on Python experience across machine learning, an LLM-based AI agent and cybersecurity. Built a course recommendation agent using the Groq LLM, a phishing URL detector with a Flask dashboard and a skin lesion classifier using EfficientNetB7 transfer learning, and completed a cybersecurity internship. Also built an Arduino-based embedded system. Seeking an entry-level role in software development, machine learning, AI or cybersecurity.

TECHNICAL SKILLS

- **Programming:** Python, Embedded C/C++, Bash, PowerShell
- **ML & AI:** TensorFlow/Keras, Scikit-learn, Pandas, NumPy, TF-IDF, Random Forest, CNN and transfer learning, LLM API integration (Groq)
- **Web & Data:** Flask, JSON
- **Cybersecurity & Cloud:** Networking fundamentals (TCP/IP, DNS, subnetting), Kali Linux, VMware, Windows Server, Active Directory, AWS EC2, Apache, Nginx
- **Tools:** Git/GitHub, Linux, VS Code
- **Embedded & Hardware:** Arduino Mega, sensor interfacing, KiCad (schematic capture and PCB layout fundamentals), Verilog basics

PROJECTS

Course Recommendation Agent | github.com/zk946190-ui/AI-agent
Python, Groq LLM API, JSON
- Built an AI agent that reads student profiles and a course catalogue (JSON) and uses the Groq LLM to generate a personalised learning path, prerequisites, timeline and career advice.

AI-Based Phishing URL Detection | github.com/zk946190-ui/phishing-url-detection
Python, Scikit-learn, Pandas, TF-IDF, Random Forest, Flask
- Trained a TF-IDF + Random Forest classifier to label URLs as phishing or safe (78.1% accuracy on 498 test URLs) and built a Flask web dashboard for real-time checks.

Skin Cancer Detection using CNN | github.com/zk946190-ui/skin-cancer-detection-cnn
Python, TensorFlow/Keras, EfficientNetB7
- Classified skin lesion images as benign or malignant; improved test accuracy from 45.5% (CNN from scratch) to 83.3% using EfficientNetB7 transfer learning.

Reverse Vending Machine (Team project, 4 members) | github.com/zk946190-ui/reverse-vending-machine
Arduino Mega, Embedded C/C++, HX711 load cell, inductive sensor, GSM, Bluetooth
- Built a recycling machine that sorts plastic, metal and glass using an inductive sensor and load cell, with keypad login, EEPROM-stored points, Bluetooth admin panel and GSM SMS alerts.
- Tested with 50 bottles per type: 90% (plastic), 100% (metal), 80% (glass) sorted correctly.

INTERNSHIP

Cybersecurity Intern — GTTC, Hubli | February 2026 – May 2026
- Studied networking fundamentals (IP addressing, subnetting, ports, protocols, OSI model) and set up Kali Linux, Windows Server and Ubuntu in VMware.
- Managed users, groups and organizational units in Active Directory; launched AWS EC2 servers and configured Apache and Nginx (hosting, reverse proxy, load balancing).
- Practised PowerShell and Bash scripting; completed a capstone project, the phishing URL detector above.

EDUCATION

B.E. in Electronics and Communication Engineering | 2022–2026
Government Engineering College, Talakal, Koppal (Visvesvaraya Technological University, Belagavi) | CGPA: 7.95/10

TRAINING & CERTIFICATIONS

- PCB Design Training — KiCad (schematic capture and PCB layout fundamentals)
- Cybersecurity and Ethical Hacking — Cisco learning certificates
- Cybersecurity internship completion certificate — GTTC, Hubli